

Evaluation Framework for Multimodal Language Models Applied to Geospatial Open Data in Agriculture

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Abstract

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1. Introduction

The rapid development of multimodal Large Language Models (LLMs) has enabled a ... ha abierto un abanico de posibilidades a nuevas aplicaciones. Entre ellas destaca el de la smart agriculture, con grandes avances en agricultura de precisión... El surgimiento de los geofoundation models [papers] y los earth embeddings [papers], así como el veloz avance de los propios modelos de lenguaje, destacando especialmente los open source, son realmente prometedores en el campo de earth observation y smart agriculture. En los últimos años han surgido una gran cantidad de estudios acerca de cómo aplicar la IA generativa al campo, pasando en muchos casos por asistentes conversacionales [papers]. Estos estudios en muchos casos utilizan RAG, geoembeddings o aplicaciones más "clásicas" como segmentación semántica sobre datos, en la mayoría de los casos, satelitales. Sin embargo, en la mayoría de los casos, las respuestas obtenidas de estas nuevas herramientas siguen siendo, aunque mucho más precisas y utilizando nuevas fronteras... genéricas. Recientemente nosotros hemos propuesto una herramienta que permite generar respuestas acerca del terreno para su aplicación a la agricultura que sean específicas al terreno AoI -> gracias a la recuperación, procesamiento y fusión de distintas fuentes multi-dominio de datos abiertas. Una vez esta herramienta ha sido validada técnicamente, es cuando surge el reto al que nos enfrentamos en este trabajo, el de evaluar las respuestas. La evaluación de las respuestas de IA generativa lleva siendo desde el origen de la misma, una de las principales líneas de investigación abiertas, ya que es común a toda aplicación que utilice inferencia no determinista. A diferencia del ML... aquí las respuestas correctas forman un espacio infinito... Existen muchas alternativas, como LLM-as-a-judge [paper], que fue el approach utilizado en [paper]. Esta técnica, aunque muy extendida gracias a su simplicidad y bajo coste operativo, is highly vulnerable to technical hallucinations, suffers from self-reference bias, and tends to lose consistency. para las aplicaciones de RAG, existen frameworks como RAGAS... tambien se ha investigado mucho las métricas que optimizan la evaluación... En nuestro caso, dado que la herramienta tiene

la particularidad de que tiene contexto amplificado con fuentes externas, ninguno de los frameworks de evaluación encontrados en la literatura se ajustan a nuestras necesidades, por lo que en este trabajo proponemos una novel metodología de evaluación para LLMs multimodales de contexto amplificado aplicados a datos de agricultura. Esta metodología es reproducible... multietapa, híbrida, mitiga sesgos etc

2. Materials and methods

2.1. System architecture

copiado y opegado del otro paper:

The proposed system is designed to combine several in terconnected modules, which integrate heterogeneous data sources. The overall architecture, included in Fig. 3, is composed of two main layers: user interface and backend processing, which includes the LLM reasoning. The aim of the data flow is to provide the final LLM with the most rich context regarding the area of interest (a specific parcel) in order for it to give the best possible answer to the user's query.

The backend consists of five different blocks, which con form a data stream from the user query to the final LLM, and then back to the user with the answer: (i) plot info retrieval, which is responsible for fetching, from the SIGPAC registry, both plot data-as slope, elevation, area, perimeter or land use , and the geo-referenced geometry. Once known the area of interest, the (ii) context retrieval block performs two parallel tasks, each leveraging different open-source data sources to enrich the context: (a) an ortophoto of the plot, obtained from aerial imagery, is processed through a multimodal LLM along with the plot data, generating a detailed description of the terrain. In addition, (b) satellite multispectral data, specifically from Sentinel-2 (Copernicus mission, European Spacial Agency) are retrieved. After spacial and temporal processing combining multiple spectral bands, an array of vegetation and water indices that reflect biophysical conditions of the land are obtained. In parallel, (iii) the system queries a document repository through a retrieval-augmented generation

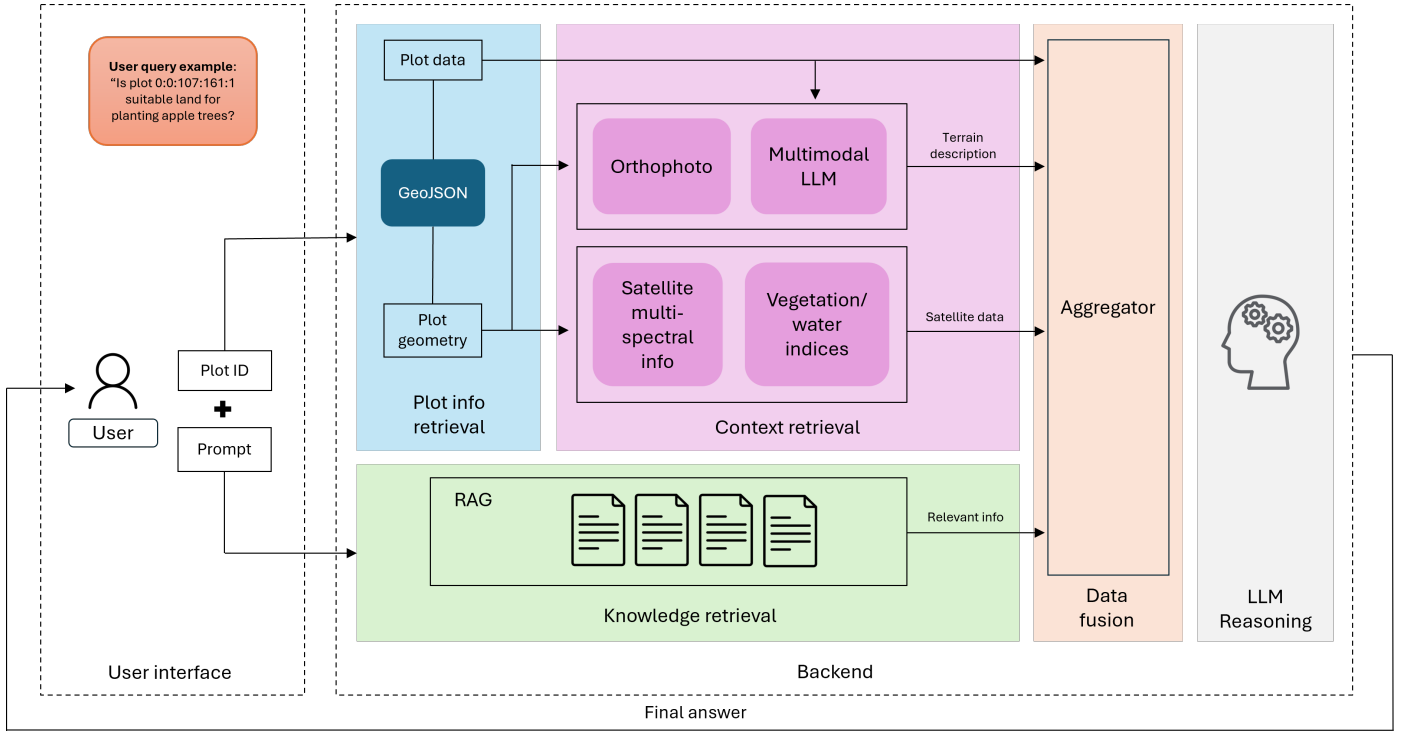


Figure 1:

mechanism, which provides relevant textual knowledge such as agronomic guidelines, regulations, or best practices. All these multi-domain data are then fused through (iv) an aggregator, which converts all data to natural language, and prepares the information into an encapsulated prompt that provides the (v) final LLM with the overall context of the plot and the query. Several open-source models were evaluated, including Qwen3-32B, Llama-3.3-70B or Mistral-Small-3.2-24B, being Qwen the one chosen for the final implementation due to its superior performance. Regarding the LLMs framework, the model inference is managed with the inference engine vLLM [9], orchestrated through the litellm proxy [10], which agglutinates all models in one port. Besides from the multimodal and final LLMs, there are also embeddings, reranker, and tool triage models working together. The infrastructure runs in a 8x80GB cluster of NVIDIA GPUs.

2.2. Evaluation framework

El principal problema que se plantea resolver, además de obtener una evaluación sistemática, rigurosa y reproducible, es el de ser capaces de evaluar no solo si la respuesta es buena, sino también en qué medida esa calidad es gracias a los módulos de su arquitectura.

Internal Knowledge over-reliance → sesgo de conocimiento interno

Therefore, **an experimental evaluation methodology has been designed**, which is detailed in this section. This approach, designed to optimize the consumption of operational resources while maintaining statistical robustness, consists of

two main stages: evaluation of the complete system and analysis of marginal contributions through an ablation study.

The main challenge of this experimental methodology, and what differentiates it from other existing approaches, is that it must be able to address the evaluation of geospatial context usage in response generation, not merely the quality of the responses themselves. A response that is correct but generic, or even seemingly specific but generated from the model's base knowledge, is not equivalent to a *personalized* response adapted to terrain characteristics. Therefore, multiple stages are combined, different configurations are evaluated, and specific metrics are proposed.

2.3. Parcel Selection and Heterogeneity

A total of 15 parcels were selected, categorized into four different Land Use (LU) types. The distribution of evaluations is weighted according to the operational utility of the tool for each functional land use, as shown in 1.

To ensure an unbiased evaluation, parcels were selected to represent a high degree of heterogeneity in their physical characteristics, including variable slopes (0.5% to 21.6%), altitudes, irregular shapes, and diversity in vegetation health (NDVI ranges). The parcels are shown in 3.

2.4. Phase I: Broad Performance Assessment

This phase evaluates the overall system capability across diverse scenarios. A total of 71 queries were designed, distributed as follows:

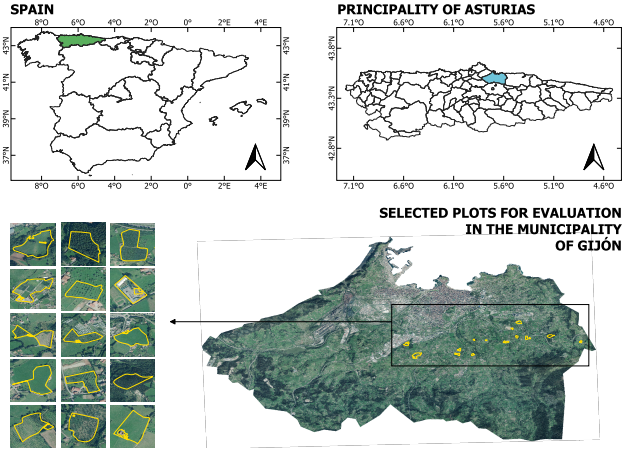


Figure 2:

Table 1: Distribution of parcels by Land Use (LU) type

Land Use	Number of queries
Fruit Trees (FR)	7
Pasture (PA)	4
Unproductive (UN)	2
Forest (FO)	2

- **Productive Parcels (FR and PA):** 5 queries per parcel, focused on irrigation, fertilization, pest management, and investment ROI.
- **Non-Productive Parcels (UN and FO):** 4 queries per parcel, consisting of 2 functional queries and 2 *Negative Testing Queries* (scenarios where the system must identify unsuitability).

2.5. Phase II: Ablation Study (Architecture Validation)

To quantify the impact and marginal contribution of each architectural module, an ablation study was conducted using a **factorial design**. Ablation studies consist of the individual analysis of different subsets (configurations) of a system, obtained through systematic activation and deactivation of its components. In this case, with three modules (Orthophoto, Satellite, and RAG, previously shown in ??), the factorial design of experiments yields $2^3 = 8$ possible configurations.

- **Configurations:** 8 modular combinations + 1 Zero-Shot baseline (the LLM without parcel context) = 9 total configurations, listed in 2.
- **Sample subset:** 1 representative parcel per Land Use category.
- **Query volume:** 2 specific queries per parcel across the 9 configurations = 72queries.

2.6. Evaluation Volume Summary

The combined results provide a total set of 143 evaluations (3). In Appendix A, a link to the dataset created for this evaluation is included, where all queries, responses, and context can be consulted.

2.7. Evaluation Subjects and Bias Mitigation

To ensure balanced validation, mitigate individual biases, and capture a wide range of perspectives, the evaluation process is carried out independently by **four differentiated evaluators**:

- **Human Evaluators (2):**
 - *Expert Evaluator:* PhD in Biology.
 - *Control Evaluator:* Non-specialist profile.
- **LLM-as-a-judge (2):**
 - *Proprietary Model (Gemini):* Gemini3.1 Flash Lite
 - *Open Source Model (Qwen):* Qwen3.5-397B-A17B-Instruct

2.8. Multidimensional Metrics Framework

Response quality is measured according to a multidimensional scale consisting of five metrics, divided into those scored considering context and those that are not, as shown in 4. Scores are given on a *Likert* scale (integer values between 1 and 5).

2.9. Preliminary Results and Discussion

This section presents some preliminary evaluation results. It should be noted that the graphs included here are only an initial overview that sheds light on the system evaluation; more advanced statistical analyses are reserved for a future scientific publication. First, 5 contains the average scores of each of the 4 evaluators across the 5 metrics, grouped by configuration. From this graph, numerous observations can be made, with the most notable being those related to evaluator correlation, and especially regarding the marginal contributions of the modules. Recalling 2, we see that configurations 2 and 3, as well as 6 and 7, include the orthophoto module; and in all metrics, a higher average score is observed in these cases, allowing us to deduce that the marginal contribution of the orthophoto module is significant.

Similarly, the trend across all metrics is ascending along the x-axis (values are lower in left-side configurations), showing how, for example, system utility in the *Zero Shot* configuration is below 1/5 for most evaluators. Since this configuration corresponds to a standard LLM model without context augmentation, and with configurations progressively ordered by active modules, it can be concluded that **responses, as expected, improve as more geospatial terrain information is added**. This result consequently verifies the initial hypothesis and validates the proposed architecture.

More generally, ?? represents the average metric values from the *Broad* stage evaluations, i.e., the complete system. For all

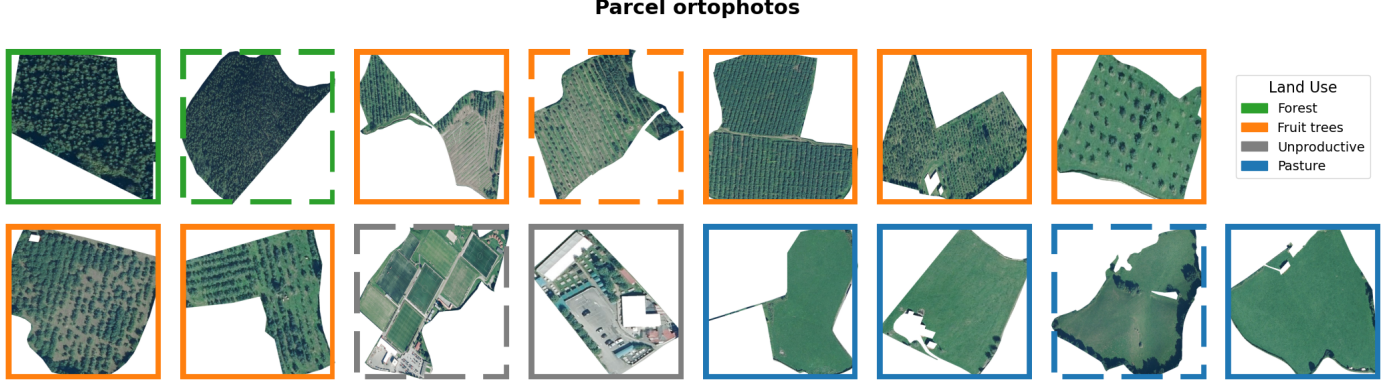


Figure 3: Selection of parcels for tool evaluation, grouped by Land Use. The number of parcels per LU depends on their functional utility in agriculture. Dashed borders correspond to parcels used in ablation study.

Table 2: Configuraciones modulares del estudio de ablación. PD means Plot Data. Active tool is always enabled because it is responsible for identifying parcel IDs as so.

Modules	RAG	ORTHO	SAT	PD	Active Tool	Code
Baseline	0	0	0	1	1	0
B + SAT	0	0	1	1	1	1
B + ORTHO	0	1	0	1	1	2
B + ORTHO + SAT	0	1	1	1	1	3
B + RAG	1	0	0	1	1	4
B + RAG + SAT	1	0	1	1	1	5
B + RAG + ORTHO	1	1	0	1	1	6
Complete	1	1	1	1	1	7
ZERO SHOT	0	0	0	0	1	ZS

RAG	Orthophoto	Satellite
Baseline Model		

Figure 4:

Table 3: Summary of total query volume in the evaluation.

Evaluation Phase	Methodology	Num. of queries
Broad Assessment	Sequential	71
Ablation Study	Simultaneous	72
Total		143

Table 4: Multidimensional evaluation metrics framework. Note: The *Grounding* metric measures the degree to which responses are derived from the obtained context, rather than generic knowledge, making it the only metric requiring context for evaluation. The *Correctness* metric does not apply to the human control evaluator.

Metric	Type
Correctness	Without Context
Completeness	Without Context
Relevance	Without Context
Utility	Without Context
Grounding	With Context

evaluators, the resulting metrics average values very close to 5/5, implying that the system has responded with a high degree

Table 5: .

Metric	H_C	H_E	LLM_Q	LLM_G
Correctness	4.690	4.704	4.817	4.746
Completeness	4.803	4.817	4.944	4.958
Relevance	4.549	4.817	4.831	4.930
Utility	4.704	4.761	4.859	4.944
Grounding	4.606	4.901	4.296	4.549

of success to the evaluated queries. These preliminary results serve as evidence that the proposed system is capable of generating high-quality, complete, useful, and generally successful responses in the broad sense.

Shapley Values

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} (v(S \cup \{i\}) - v(S)) \quad (1)$$

N is the set of all players: $N = \{M_S, M_O, M_R\}$ and S is any subset or *coalition*.

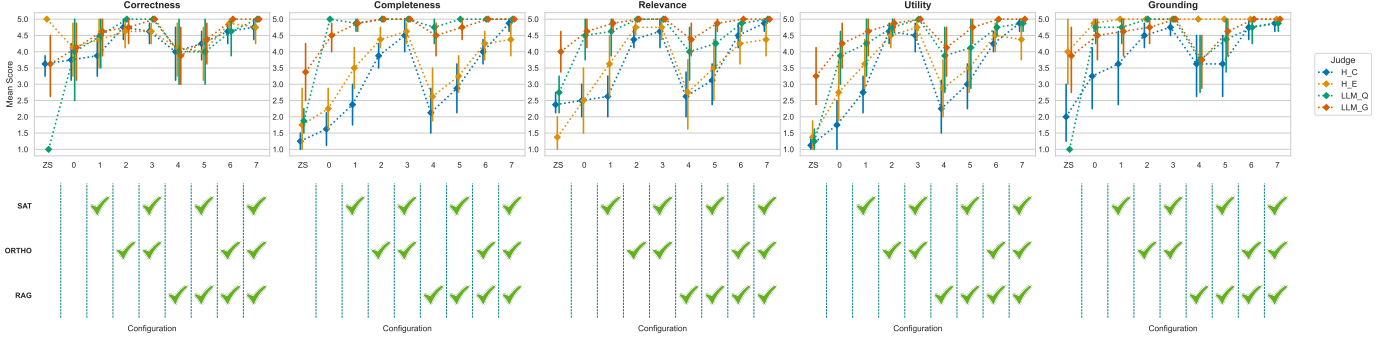


Figure 5: Preliminary evaluation results from the *Ablation Study* phase for the 4 evaluators and 5 metrics grouped by configuration. Differences can be observed based on active modules.

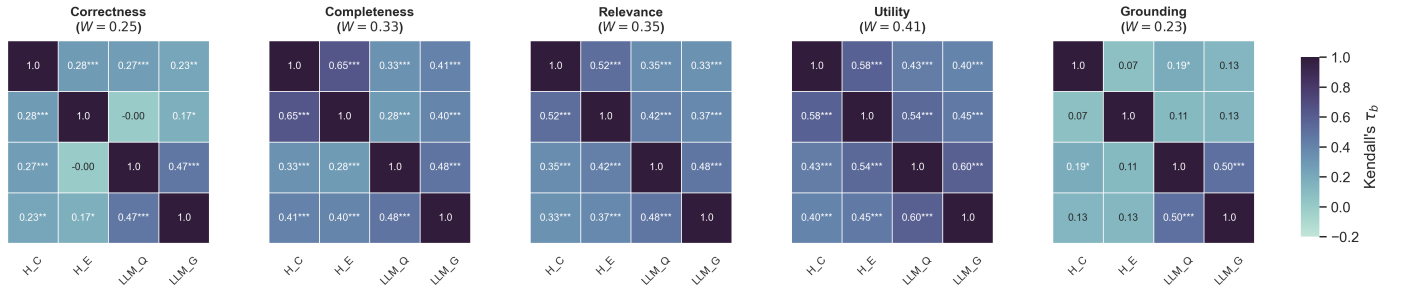


Figure 6:

$$\begin{aligned}
 \phi_S(v) = & \frac{1}{3}[v(\{M_S\}) - v(\emptyset)] \\
 & + \frac{1}{6}[v(\{M_S, M_O\}) - v(\{M_O\})] \\
 & + \frac{1}{6}[v(\{M_S, M_R\}) - v(\{M_R\})] \\
 & + \frac{1}{3}[v(\{M_S, M_O, M_R\}) - v(\{M_O, M_R\})]
 \end{aligned} \tag{2}$$

$$\begin{aligned}
 \phi_O(v) = & \frac{1}{3}[v(\{M_O\}) - v(\emptyset)] \\
 & + \frac{1}{6}[v(\{M_S, M_O\}) - v(\{M_S\})] \\
 & + \frac{1}{6}[v(\{M_O, M_R\}) - v(\{M_R\})] \\
 & + \frac{1}{3}[v(\{M_S, M_O, M_R\}) - v(\{M_S, M_R\})]
 \end{aligned} \tag{3}$$

$$\begin{aligned}
 \phi_R(v) = & \frac{1}{3}[v(\{M_R\}) - v(\emptyset)] \\
 & + \frac{1}{6}[v(\{M_S, M_R\}) - v(\{M_S\})] \\
 & + \frac{1}{6}[v(\{M_O, M_R\}) - v(\{M_O\})] \\
 & + \frac{1}{3}[v(\{M_S, M_O, M_R\}) - v(\{M_S, M_O\})]
 \end{aligned} \tag{4}$$

$$\begin{aligned}
 \phi_S(v) = & \frac{1}{3}[v(C_1) - v(C_0)] + \frac{1}{6}[v(C_3) - v(C_2)] \\
 & + \frac{1}{6}[v(C_5) - v(C_4)] + \frac{1}{3}[v(C_7) - v(C_6)]
 \end{aligned} \tag{5}$$

$$\begin{aligned}
 \phi_O(v) = & \frac{1}{3}[v(C_2) - v(C_0)] + \frac{1}{6}[v(C_3) - v(C_1)] \\
 & + \frac{1}{6}[v(C_6) - v(C_4)] + \frac{1}{3}[v(C_7) - v(C_5)]
 \end{aligned} \tag{6}$$

$$\begin{aligned}
 \phi_R(v) = & \frac{1}{3}[v(C_4) - v(C_0)] + \frac{1}{6}[v(C_5) - v(C_1)] \\
 & + \frac{1}{6}[v(C_6) - v(C_2)] + \frac{1}{3}[v(C_7) - v(C_3)]
 \end{aligned} \tag{7}$$

Meter gráficas solo con valores del promedio de los 4 jueces para no crear ruido visual

3. Results and Discussion

Referencia a zenodo con dataset anonimizado

4. Conclusions

Acknowledgements

Thanks to ...

Appendix A. Appendix A: Shapley values

References

Table 6: Shapley

Judge	Metric	Module	Shapley Value	Relative Contribution (%)	Global Delta
H_C	Correctness	R	0.167	16.67%	1.000
		O	0.729	72.92%	
		S	0.104	10.42%	
	Completeness	R	0.396	12.18%	3.250
		O	2.083	64.10%	
		S	0.771	23.72%	
	Relevance	R	0.229	9.65%	2.375
		O	1.854	78.07%	
		S	0.292	12.28%	
	Utility	R	0.271	8.67%	3.125
		O	2.208	70.67%	
		S	0.646	20.67%	
	Grounding	R	0.208	12.82%	1.625
		O	1.208	74.36%	
		S	0.208	12.82%	
H_E	Correctness	R	0.021	3.33%	0.625
		O	0.583	93.33%	
		S	0.021	3.33%	
	Completeness	R	-0.021	-0.98%	2.125
		O	1.541	72.55%	
		S	0.604	28.43%	
	Relevance	R	-0.146	-7.78%	1.875
		O	1.479	78.89%	
		S	0.542	28.89%	
	Utility	R	-0.104	-6.41%	1.625
		O	1.333	82.05%	
		S	0.396	24.36%	
	Grounding	R	0.042	33.33%	0.125
		O	0.042	33.33%	
		S	0.042	33.33%	
LLM_Q	Correctness	R	-0.146	-14.58%	1.000
		O	0.854	85.42%	
		S	0.292	29.17%	
	Completeness	R	-0.062	0.00%	0.000
		O	0.062	0.00%	
		S	0.000	0.00%	
	Relevance	R	-0.250	-50.00%	0.500
		O	0.625	125.00%	
		S	0.125	25.00%	
	Utility	R	-0.062	-6.25%	1.000
		O	0.812	81.25%	
		S	0.250	25.00%	
	Grounding	R	-0.479	-383.33%	0.125
		O	0.458	366.67%	
		S	0.146	116.67%	
LLM_G	Correctness	R	-0.083	-9.52%	0.875
		O	0.667	76.19%	
		S	0.292	33.33%	
	Completeness	R	-0.021	-4.17%	0.500
		O	0.354	70.83%	
		S	0.167	33.33%	
	Relevance	R	-0.083	-22.22%	0.375
		O	0.292	77.78%	
		S	0.167	44.44%	
	Utility	R	0.000	0.00%	0.750
		O	0.500	66.67%	
		S	0.250	33.33%	
	Grounding	R	-0.208	-41.67%	0.500
		O	0.479	95.83%	
		S	0.229	45.83%	

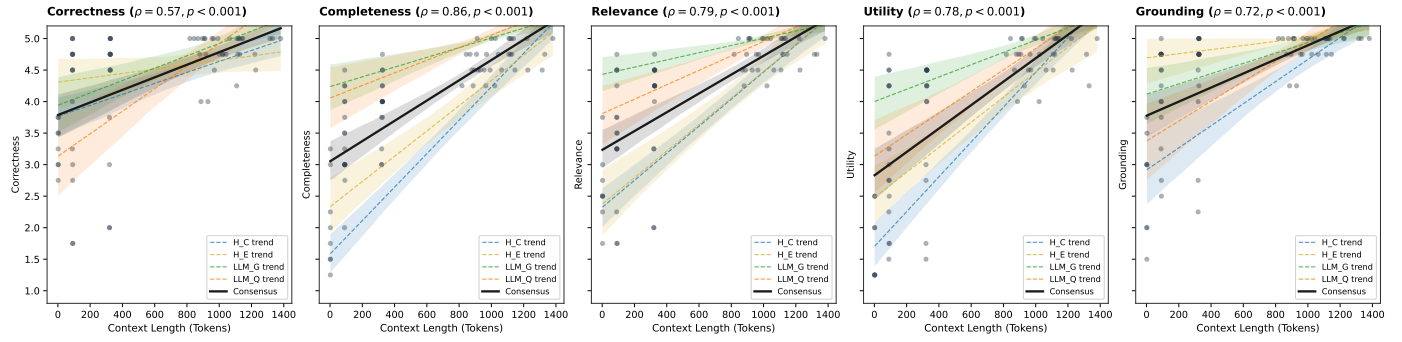


Figure 7:

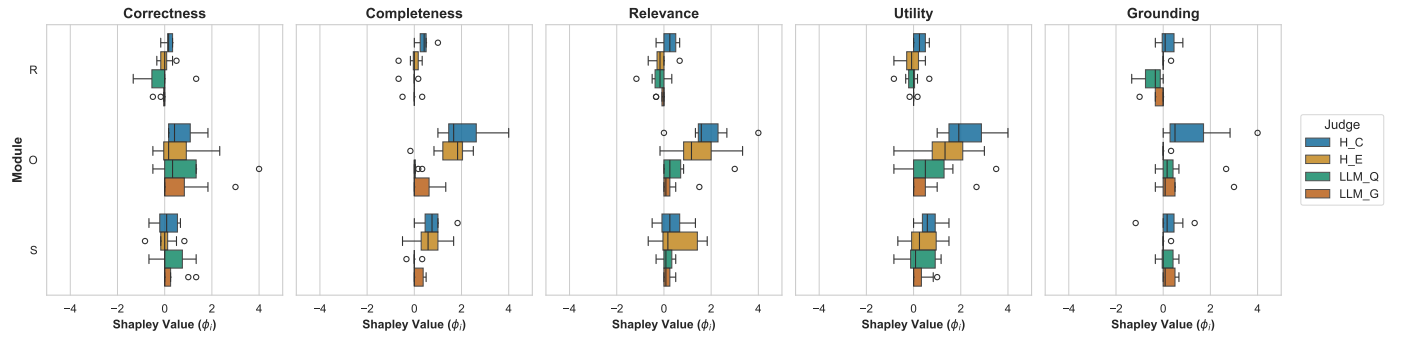


Figure 8:

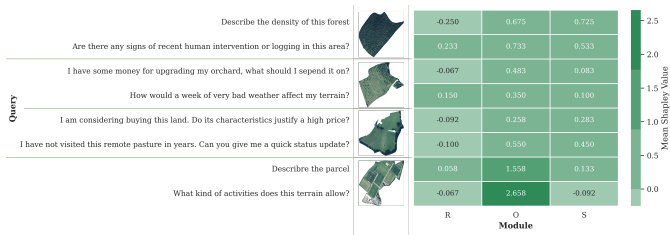


Figure 9: